

FINN WOLFF

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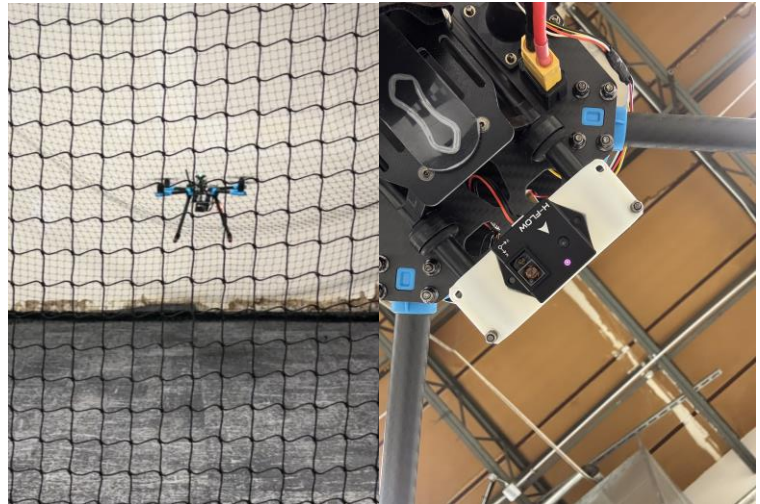
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AUTONOMOUS DRONE NAVIGATION

GPS-denied indoor flight with companion computer

Goal: Create a testbed to train and evaluate AI vision models for drone applications in a controlled environment.

- Installed optical flow sensor to replace GPS input in the EKF, making all positioning relative to launch point, allowing for safe and locally accurate navigation within netted area.
- Integrated Raspberry Pi 5 for external compute and full navigation control to send and receive instructions from laptop while in flight, allowing for highly customized missions that can react to the environment.
- Wrote script to automate endurance testing of different battery and motor combinations to determine most effective configuration for specific applications.



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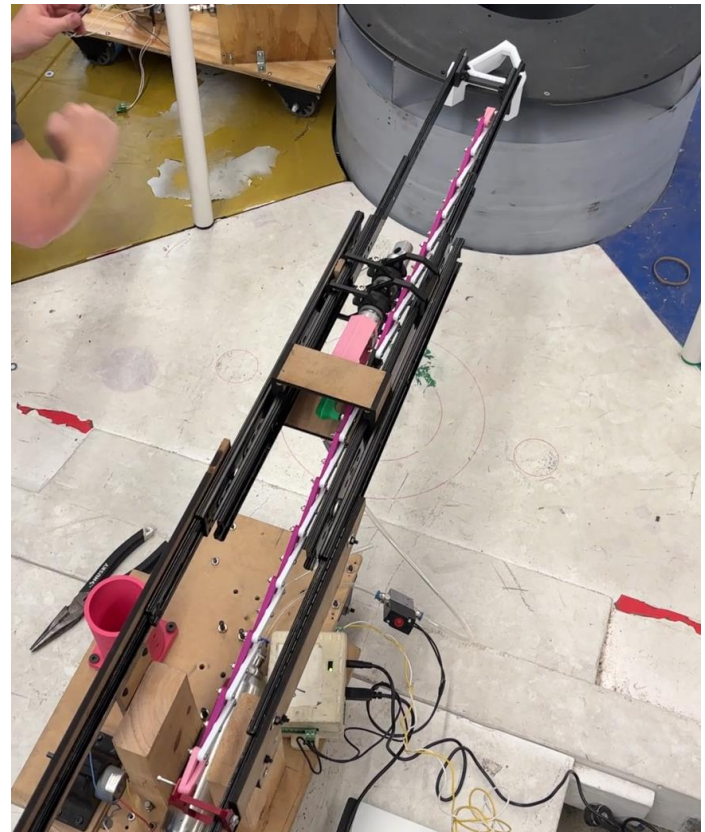
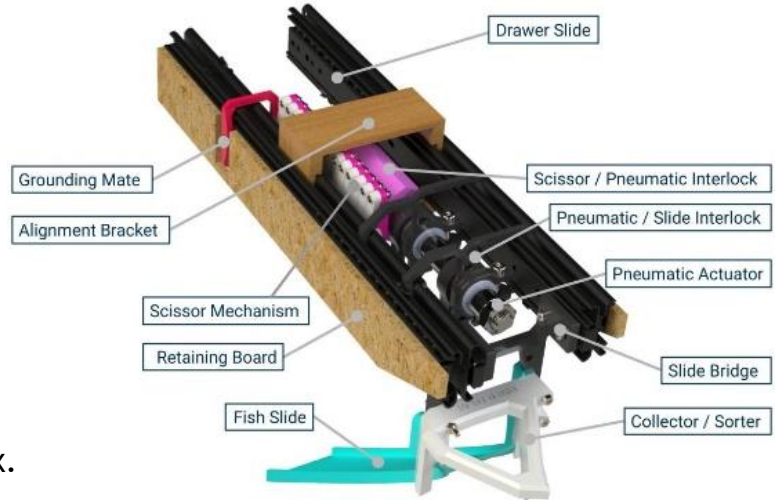
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ROBOTICS DESIGN COMPETITION

Pneumatic actuated extendable platform

Goal: Design a carriage to extend 2.5X original length in under 0.5s while carrying 5lbs reliably using a singular pneumatic actuator and automatically return to the starting position.

- Created a low friction, low tolerance, and high strength 3D printed scissor mechanism to extend actuation distance by 37x.
- Designed mechanism specifically to prevent mechanical binding across 40+ scissor links.
- Utilized low-cost off the shelf components such as drawer slides to hold high mass components.



Method	Decision Criteria								Total	Ranking
	Safe	Cost Effective	Completes Task	Constructability	Visual Appeal	Consistency	Usability	Weight (%)		
Leave Stable 2WD + Guiding Wheel	5	3	4	4	3	4	4	3.92	1	
Leave Stable 4WD	5	1	4	2	4	5	4	3.85	2	
Leave Stable 2WD + Flat Plate	5	4	3	4	2	2	3	3	3	
Extinguish Fire Extending Rail System	4	3	4	4	3	4	2	3.57	2	
Extinguish Fire Pantograph	5	3	3	2	4	2	2	2.64	3	
Extinguish Fire Tubes/Cylinders	4	4	4	4	3	4	2	3.64	1	
Grab Lever Hook	5	5	3	5	3	2	1	3	1	
Grab Lever Claw	4	3	4	1	4	3	2	2.92	2	
Refill Feeding Station Extending Rail System	4	3	4	4	3	4	2	3.57	1	
Refill Feeding Station Pantograph	5	3	2	2	4	2	2	2.42	3	
Refill Feeding Station Tubes/Cylinders	4	4	3	4	3	4	2	3.42	2	
Grab Fish and Eels Claw	4	2	3	1	4	1	3	2.21	3	
Grab Fish and Eels Wedge	5	5	4	5	3	4	1	3.78	2	
Grab Fish and Eels Cage with Hole	4	4	5	4	4	5	1	4.07	1	
Sort Fish and Eels Different Size Holes	5	5	4	4	3	2	1	3.07	2	
Sort Fish and Eels Converging Ramp	5	4	4	3	4	4	1	3.5	1	
Place Fish and Eels Tube System	5	4	4	3	5	3	1	3.28	1	
Place Fish and Eels Funnel	5	4	4	4	3	2	1	3	2	
Light Bonfire Claws	4	2	3	1	4	2	4	2.64	3	
Light Bonfire Plow	5	4	4	5	2	3	1	3.35	2	
Light Bonfire Cup	5	3	4	4	3	4	2	3.64	1	

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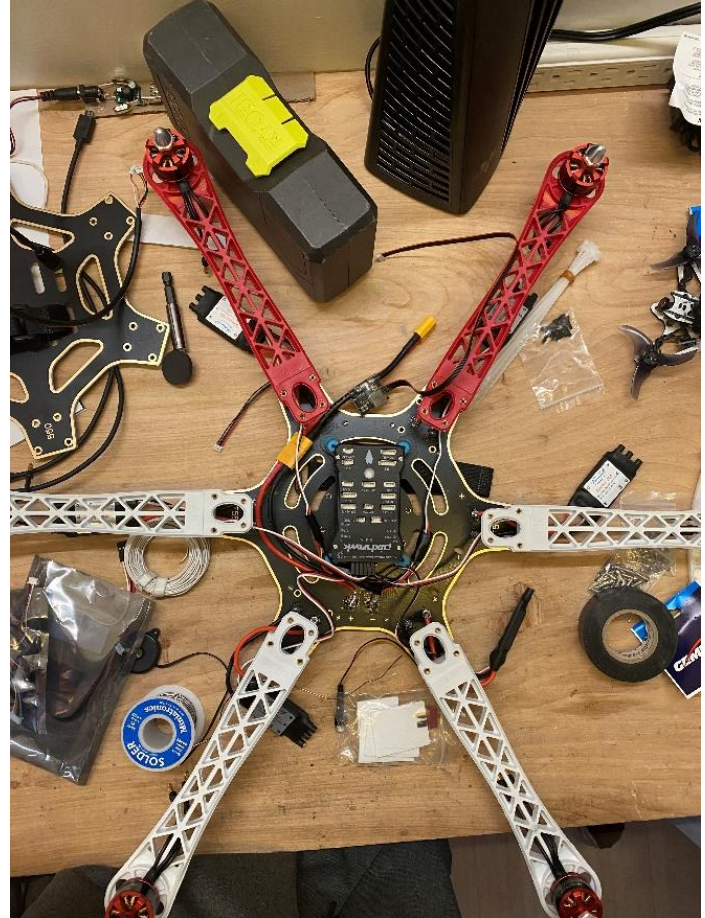
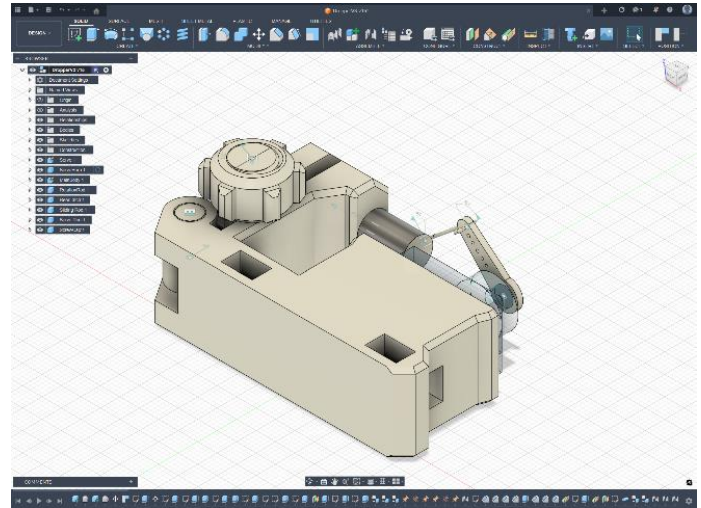
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AUTONOMOUS DELIVERY DRONE

Goal: Deploy a modified, low-cost delivery drone based off a consumer platform to transport small packages for small business within local community.

- Assembled and configured drone platform to match estimated payload weight with motor performance and propellor size.
- Designed pin-slot mechanism for package release in Fusion360 with operator safety in mind.
- Used open-source software for manual mission planning and compensated for poor satellite data with onboard collision sensors.

Hexacopter with servo actuated release hook



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PRECISION ROCKET CONTROL

Modular rocket with active control airbrake system

Goal: Design a fully 3D printable rocket body with a modular part interchange system and incorporate an airbrake system to apogee at an exact specified altitude repeatedly.

- Designed 4-layer PCB with 40+ components for main flight computer and power distribution board, reducing weight by 70g.
- Used novel real-time CFD to determine aerodynamic performance of rocket with airbrakes extended and estimate CFD.
- Created high tolerance resin printed gearing system to open airbrakes with a servo motor, allowing for quick, repeated opening while being extremely light.

